

Ensemble Quantum Computing Laboratory Prelab 2

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January 27, 2025

Problem 1

Let's first analytically calculate the fourier transform of the given function $S(t) = e^{i\Omega t} e^{i\phi} e^{-\frac{t}{T_2}}$. The fourier transform is done as follows:

$$\begin{aligned}
 \tilde{S}(\omega) &= \int_0^{+\infty} e^{i\Omega t} e^{i\phi} e^{-\frac{t}{T_2}} e^{-i\omega t} dt \\
 &= e^{i\phi} \int_0^{+\infty} e^{i(\Omega-\omega)t} e^{-\frac{t}{T_2}} dt \\
 &= e^{i\phi} \int_0^{+\infty} e^{i(\Omega-\omega)t - \frac{t}{T_2}} dt \\
 &= e^{i\phi} \int_0^{+\infty} e^{t(i(\Omega-\omega) - \frac{1}{T_2})} dt \\
 &= e^{i\phi} \frac{1}{\frac{1}{T_2} - i\Omega + i\omega}
 \end{aligned}$$

Therefore, the fourier transform of the given function is $\tilde{S}(\omega) = e^{i\phi} \frac{1}{\frac{1}{T_2} - i\Omega + i\omega}$. And the real and imaginary parts are respectively:

$$\begin{aligned}
 R &= \frac{T_2 \cos(\phi) - T_2^2 (\omega - \Omega) \sin(\phi)}{1 + T_2^2 (\omega - \Omega)^2} \\
 I &= \frac{T_2 \sin(\phi) + T_2^2 (\omega - \Omega) \cos(\phi)}{1 + T_2^2 (\omega - \Omega)^2}
 \end{aligned}$$

It is easy to see (with a little trigonometry) that $R \cos(\phi) + I \sin(\phi) = \frac{T_2}{1 + T_2^2 (\omega - \Omega)^2} = \frac{\frac{1}{T_2}}{\frac{1}{T_2^2} + (\omega - \Omega)^2}$, which is a lorentzian function. Therefore such spectrum has a pure absorption lineshape.

A plot for the real and imaginary parts of the spectrum of several ϕ is shown below in figure 1.

Problem 2

An intuitive way of understanding why reducing acquisition time to something similar to the duration of signal, some nT_2 where $n \in [2, 5]$ in this case if I'm not mistaken, or adding a weight reducing along time would increase the SNR would be to think in terms of energy. The energy of

our signal in this case is finite while the energy of the noise is infinite. Notice how the energy of the signal is concentrated in the first several T_2 and the energy of the noise is distributed uniformly along the time. Therefore, the SNR would be higher if we are more focused on the first several T_2 of the signal, either by truncating or by adding a time-anti-correlated weight.

Demonstration effort are made to show these methods works. Figure 2 shows the relation between SNR and window sizes. Figure 3 shows the relation between the weight function and SNR, the weight function here is the power of the function of the envelope of the original function, the weight index is the power.

Problem 3

The sketch is shown in Figure4

Problem 4

The full Hamiltonian is written as $H_{IS} = \Omega_I I_Z + \Omega_S S_Z + 2\pi J(I_X S_X + I_Y S_Y + I_Z S_Z)$:

$$\begin{pmatrix} 0.5\Omega_I + 0.5\Omega_S + 0.5\pi J & 0 & 0 & 0 \\ 0 & 0.5\Omega_I - 0.5\Omega_S - 0.5\pi J & \pi J & 0 \\ 0 & \pi J & -0.5\Omega_I + 0.5\Omega_S - 0.5\pi J & 0 \\ 0 & 0 & 0 & -0.5\Omega_I - 0.5\Omega_S + 0.5\pi J \end{pmatrix}$$

The diagonal elements comes from $I_Z S_Z$ and Zeeman terms, i.e. $\Omega_I I_Z + \Omega_S S_Z$. The off-diagonal elements are originated from $I_X S_X + I_Y S_Y$, terms that connects, assuming the four states are $|00\rangle$ $|01\rangle$ $|10\rangle$ $|11\rangle$, $|01\rangle$ and $|10\rangle$ states.

The logitudinal approximation is valid when $|\Omega_I - \Omega_S| \gg 2\pi J$. Under this assumption, terms $I_X S_X$ and $I_Y S_Y$ are dropped.

Problem 5

Notice for any operator A , we have: $\frac{dA}{dt} = \frac{i[H, A]}{\hbar}$ here $H = H_J = J I_z S_z$ (The matrices are just not worth writing out one by one, I did the calculation and they are all zero)

1. $A = I_X S_X: \frac{dA}{dt} = \frac{i}{\hbar} J (I_Z S_Z I_X S_X - I_X S_X I_Z S_Z) = 0$
2. $A = I_X S_Y: \frac{dA}{dt} = \frac{i}{\hbar} J (I_Z S_Z I_X S_Y - I_X S_Y I_Z S_Z) = 0$
3. $A = I_Y S_X: \frac{dA}{dt} = \frac{i}{\hbar} J (I_Z S_Z I_Y S_X - I_Y S_X I_Z S_Z) = 0$
4. $A = I_Y S_Y: \frac{dA}{dt} = \frac{i}{\hbar} J (I_Z S_Z I_Y S_Y - I_Y S_Y I_Z S_Z) = 0$

Now we know all these operators commute with $H_J = J I_z S_z$. Therefore no evolution is caused by these operators in logitudinal coupling.

Problem 6

The Operator $I_X S_X$ can be expressed by Paulies $\rho = \sigma_X \otimes \sigma_X$ accurate up to some scalar(which we will find trivial later). Similarly $I_X = \sigma_X \otimes I$ and $S_X = I \otimes \sigma_X$.

The expectation value of I_X in $I_X S_X$ is $Tr(I_X \rho) = TR(\sigma_X \sigma_X \otimes I \sigma_X) = Tr(I)Tr(\sigma_X) = 0$.
The expectation value of S_X in $I_X S_X$ is $Tr(S_X \rho) = TR(I \sigma_X \otimes \sigma_X \sigma_X) = Tr(\sigma_X)Tr(I) = 0$.
Therefore there's no measureable magnetization in this state.

Notice similar property for $I_Z S_Z$ and $I_Y S_Y$, these two states also have no measureable magnetization.

Problem 7

Problem 7 is done in a .ipynb file and hence is displayed separately below

prob7

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```
[29]: import numpy as np
import matplotlib.pyplot as plt
from qutip import tensor, qeye, sigmax, sigmaz, Qobj, mesolve

sx = sigmax()
sz = sigmaz()
E = qeye(2)
Ix = 0.5 * tensor(sx, E)
Iz = 0.5 * tensor(sz, E)
Sx = 0.5 * tensor(E, sx)
Sz = 0.5 * tensor(E, sz)

# a
rho0 = 0.25 * tensor(E, E) + 0.5 * Iz + 0.5 * Sz + Iz * Sz

print("Initial density matrix rho(0):")
print(rho0)

theta = np.pi / 2
U_I = (-1j * theta * sx / 2).expm()
U = tensor(U_I, E)
rho_rotated = U * rho0 * U.dag()

print("\nDensity matrix after /2 x-rotation on spin I (rho_rotated):")
print(rho_rotated)

# b
J = 200
H_IS = 2 * np.pi * J * Iz * Sz
t = 0.005
U_t = (-1j * H_IS * t).expm()
rho_t = U_t * rho_rotated * U_t.dag()
print(f"\nDensity matrix rho(t) at t = {t} seconds:")
print(rho_t)
print("\nInteraction Hamiltonian H_IS:")
print(H_IS)
```

```

# c
O_I = Ix
O_S = Sx

observables = [O_I, O_S]

t_max = 0.02
num_steps = 200
tlist = np.linspace(0, t_max, num_steps)

result = mesolve(H_IS, rho_rotated, tlist, [], observables)

expect_OI = result.expect[0]
expect_OS = result.expect[1]

plt.figure(figsize=(10, 6))
plt.plot(tlist * 1000, expect_OI, label=r'$\langle I_x \rangle$', color='blue')
plt.plot(tlist * 1000, expect_OS, label=r'$\langle S_x \rangle$', color='red')
plt.title('Observable Signals')
plt.xlabel('Time')
plt.ylabel('Expectation Values')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

Initial density matrix $\rho(0)$:

Quantum object: dims=[[2, 2], [2, 2]], shape=(4, 4), type='oper', dtype=CSR, isherm=True

Qobj data =

```

[[1. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]

```

Density matrix after $/2$ x-rotation on spin I (ρ_{rotated}):

Quantum object: dims=[[2, 2], [2, 2]], shape=(4, 4), type='oper', dtype=Dense, isherm=True

Qobj data =

```

[[0.5+0.j 0. +0.j 0. +0.5j 0. +0.j ]
 [0. +0.j 0. +0.j 0. +0.j 0. +0.j ]
 [0. -0.5j 0. +0.j 0.5+0.j 0. +0.j ]
 [0. +0.j 0. +0.j 0. +0.j 0. +0.j ]]

```

Density matrix $\rho(t)$ at $t = 0.005$ seconds:

Quantum object: dims=[[2, 2], [2, 2]], shape=(4, 4), type='oper', dtype=Dense, isherm=True

```

Qobj data =
[[0.5+0.j 0. +0.j 0. -0.5j 0. +0.j ]
 [0. +0.j 0. +0.j 0. +0.j 0. +0.j ]
 [0. +0.5j 0. +0.j 0.5+0.j 0. +0.j ]
 [0. +0.j 0. +0.j 0. +0.j 0. +0.j ]]

```

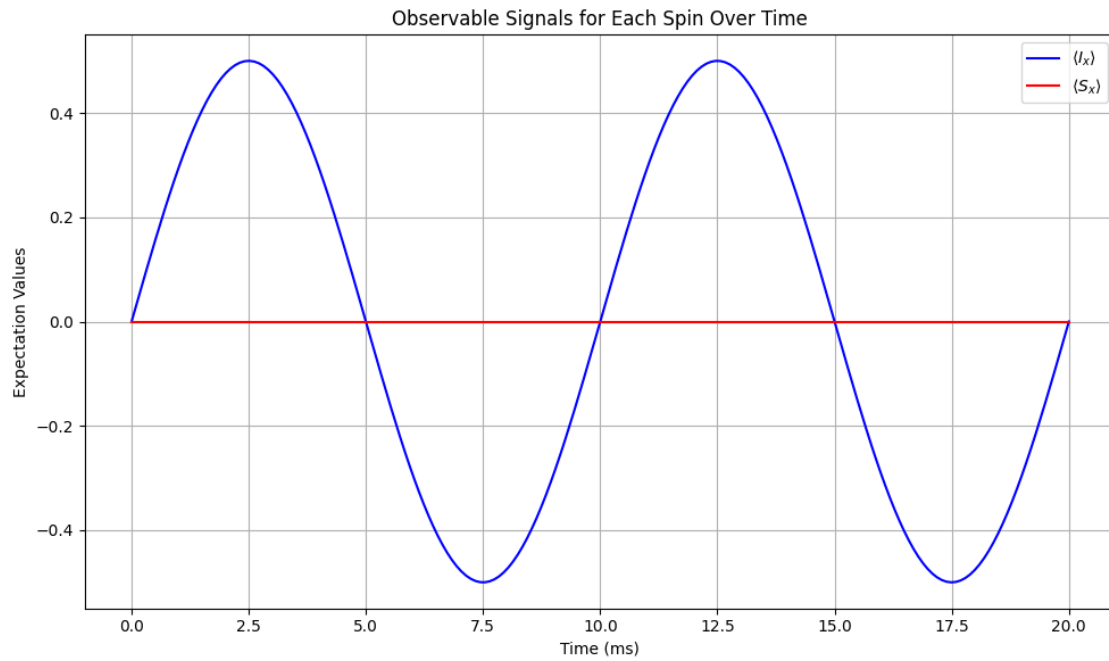
Interaction Hamiltonian H_IS:

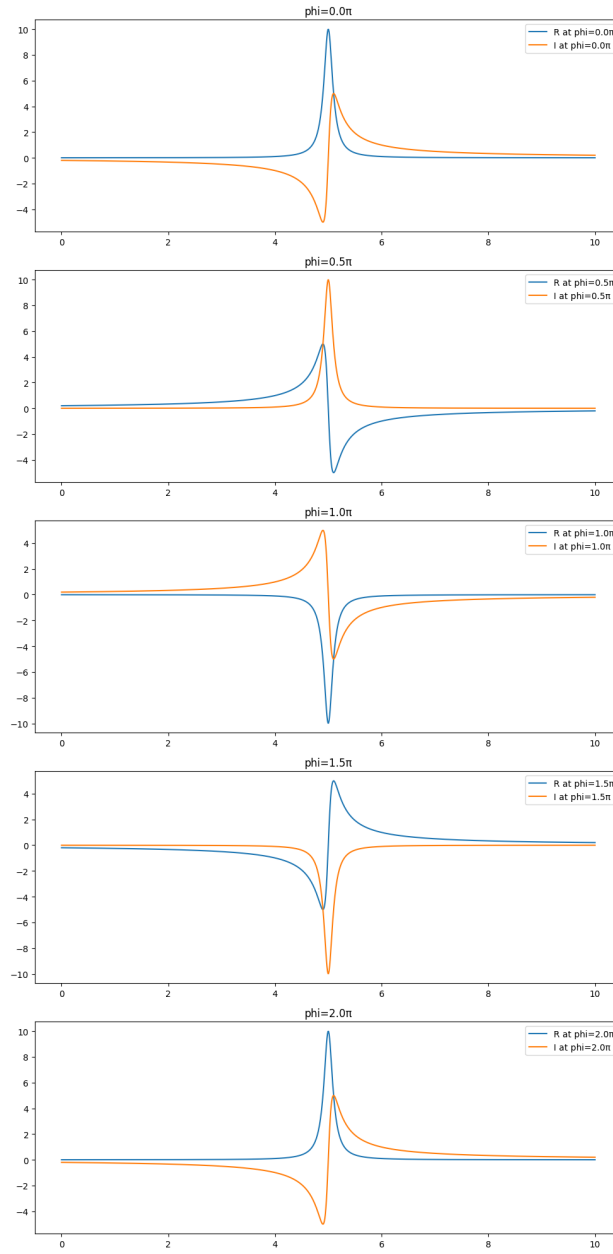
Quantum object: dims=[[2, 2], [2, 2]], shape=(4, 4), type='oper', dtype=CSR, isherm=True

```

Qobj data =
[[ 314.15926536    0.          0.          0.          ]
 [    0.        -314.15926536    0.          0.          ]
 [    0.          0.        -314.15926536    0.          ]
 [    0.          0.          0.          314.15926536]]

```



Figure 1: Plots for various valudes of ϕ

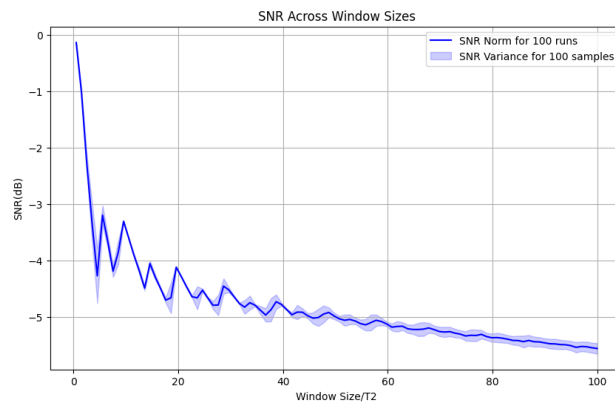


Figure 2: the relation between SNR and window size

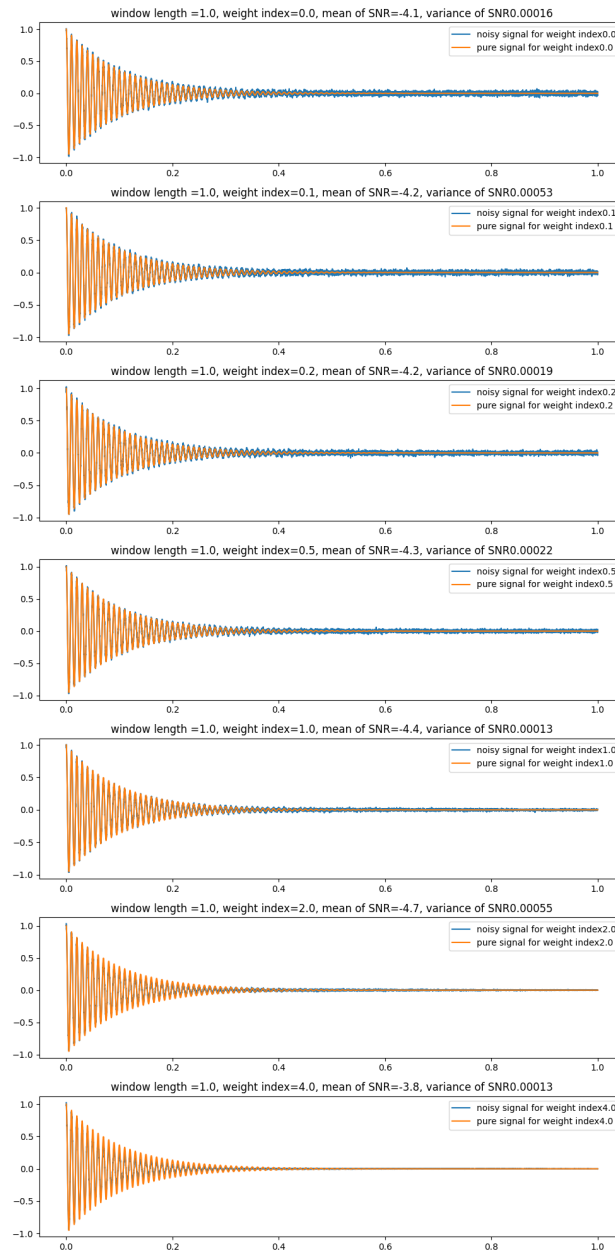


Figure 3: the relation between SNR and weight function

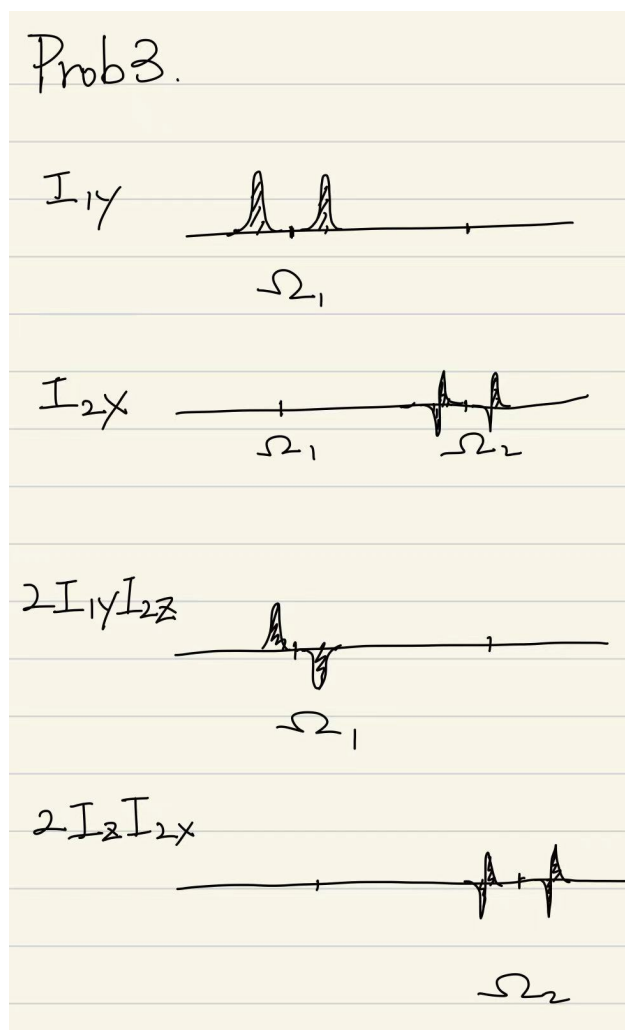


Figure 4: the sketch of spectra arising from given operators